

Miscellaneous Capital Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Miscellaneous Capital Intensity (MCI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on MCI achieves an annualized gross (net) Sharpe ratio of 0.30 (0.27), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 29 (26) bps/month with a t-statistic of 4.02 (3.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Statman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market) is 23 bps/month with a t-statistic of 3.60.

1 Introduction

Asset prices reflect the fundamental production decisions of firms operating in competitive markets with technological and financial frictions. The cross-section of expected returns emerges from heterogeneity in firms' exposure to systematic productivity shocks and their differential abilities to adjust capital stocks in response to changing economic conditions. Despite extensive theoretical work linking production characteristics to risk premia through investment-based asset pricing models, empirical evidence connecting specific measures of capital intensity to expected returns remains incomplete. We identify and test a novel predictor of stock returns based on miscellaneous capital intensity (MCI), which captures firms' deployment of specialized productive assets not classified in standard capital categories. This measure provides unique information about firms' production technologies and their exposure to aggregate productivity risk that existing capital measures fail to capture.

Production-based asset pricing theory predicts that firms with higher marginal costs of production earn higher expected returns to compensate investors for bearing systematic risk [Cochrane and Piazzesi \(2005\)](#). In a production economy with capital adjustment costs, firms' optimal investment policies depend on the interaction between productivity shocks and the convexity of adjustment cost functions [Zhang \(2005\)](#). Miscellaneous capital intensity captures firms' reliance on specialized, difficult-to-adjust productive assets that amplify exposure to aggregate productivity shocks through both operating and financial leverage channels [Carlson et al. \(2004\)](#). When productivity declines, firms with high miscellaneous capital intensity face steeper increases in marginal production costs due to the quasi-fixed nature of specialized capital, creating procyclical conditional risk premia.

The economic mechanism linking MCI to expected returns operates through the interaction of irreversibility constraints and time-varying risk premia in production economies [Cooper and Priestley \(2009\)](#). Firms with substantial miscellaneous capital

commitments cannot easily redeploy these assets during economic downturns, leading to higher systematic risk exposure through the operating leverage channel. This effect intensifies when capital adjustment costs are convex, as theoretical models predict that firms with less flexible production technologies require higher risk compensation [Berk et al. \(1999\)](#). The cross-sectional variation in MCI thus proxies for differences in firms' conditional betas with respect to aggregate productivity shocks.

Production-based models with heterogeneous capital goods predict that the composition of firms' capital stocks determines their risk profiles through differential exposure to technology shocks [Kogan and Papanikolaou \(2014\)](#). Miscellaneous capital represents investments in specialized equipment, proprietary systems, and unique productive assets that embody firm-specific technological choices. These investments create real options that are particularly sensitive to aggregate uncertainty, as the value of growth options embedded in specialized capital depends critically on expected productivity growth [Ai and Kiku \(2013\)](#). Consequently, cross-sectional differences in MCI should predict returns through their correlation with firms' loadings on production-based risk factors that price the cross-section of equity returns.

We find strong evidence that miscellaneous capital intensity predicts cross-sectional variation in expected returns. A value-weighted long/short trading strategy based on MCI achieves an annualized gross Sharpe ratio of 0.30 and delivers a monthly average abnormal return of 29 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 4.02. The strategy's performance remains robust after accounting for transaction costs, generating a net Sharpe ratio of 0.27 and a net alpha of 26 basis points per month with a t-statistic of 3.74. These results indicate that MCI captures a distinct source of systematic risk not spanned by existing factor models.

The predictive power of MCI extends across different segments of the equity market and remains significant among the largest stocks. Within the quintile of

firms with market capitalizations above the 80th NYSE percentile, the MCI strategy generates an average return of 21 basis points per month with a t-statistic of 2.26. The strategy’s alpha relative to the six-factor model equals 33 basis points per month among these large-cap stocks, with a t-statistic of 3.94. Furthermore, controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the MCI strategy maintains a significant alpha of 23 basis points per month with a t-statistic of 3.60.

The economic magnitude of the MCI premium places it among the more robust predictors in the anomaly literature. The strategy’s gross Sharpe ratio exceeds 65% of documented anomalies, while its net Sharpe ratio surpasses 86% of existing strategies after accounting for implementation costs. A dollar invested in the MCI strategy would have grown to \$2.93 on a gross basis and \$2.31 net of transaction costs, ranking in the top 10% and top 6% of anomaly strategies, respectively. These results demonstrate that MCI contains unique information about expected returns that significantly expands the investment opportunity set available to investors.

Our study contributes to the production-based asset pricing literature by providing novel empirical evidence linking a specific measure of capital intensity to the cross-section of expected returns. While [Fama and French \(1992\)](#) and [Berk et al. \(1999\)](#) establish theoretical connections between book-to-market ratios and risk through investment dynamics, and [Novy-Marx \(2011\)](#) demonstrates the pricing relevance of operating leverage, we identify miscellaneous capital intensity as a distinct characteristic that captures firms’ exposure to productivity risk through specialized capital investments. Our findings extend the empirical support for production-based theories of asset pricing beyond traditional measures of capital structure and operating efficiency documented in the *Journal of Financial Economics* and *Review of Financial Studies*.

Methodologically, we advance the anomaly testing literature by implementing the

comprehensive protocol of [Novy-Marx and Velikov \(2023b\)](#), which addresses concerns about data mining and implementation costs that plague many cross-sectional predictors. Unlike studies in the *Journal of Accounting Research* that focus on accounting-based measures of profitability, we demonstrate that MCI’s predictive power survives stringent robustness tests including controls for market capitalization, alternative portfolio construction methods, and spanning tests against closely related anomalies. Our use of high-frequency effective spread estimates to calculate implementation costs provides more accurate assessments of strategy profitability than approaches using simple bid-ask spreads common in earlier *Journal of Finance* studies.

The broader implications of our findings extend to both theoretical and practical applications in asset pricing. For theorists, the success of MCI as a return predictor suggests that models incorporating heterogeneous capital goods and specialized production technologies, as explored in recent *Journal of Accounting and Economics* papers, merit further development. For practitioners, the MCI strategy’s ability to generate significant net alphas after controlling for existing factors and transaction costs indicates an economically meaningful investment opportunity. The strategy’s robust performance across different market segments and time periods, combined with its ability to improve mean-variance efficient frontiers even in the presence of the factor zoo, establishes miscellaneous capital intensity as an important characteristic for understanding cross-sectional variation in expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several

decades of financial reporting. This database offers standardized accounting variables that enable consistent measurement across firms and over time. Miscellaneous Capital Intensity signal combines two key accounting variables. Current assets - other sundry (ACOX) represents miscellaneous current assets not classified in standard categories such as cash, receivables, or inventory. This item captures various short-term assets including prepaid expenses, deferred charges, and other current assets that firms report but do not itemize separately. Invested capital (ICAPT) measures the total capital deployed in a firm’s operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable represents the permanent capital base that stakeholders have committed to the firm. We construct the Miscellaneous Capital Intensity signal by dividing current assets - other sundry (ACOX) by invested capital (ICAPT). This ratio captures the proportion of a firm’s permanent capital base allocated to miscellaneous current assets, providing insight into the firm’s working capital management and asset composition. We calculate this measure annually using fiscal year-end values from firms’ financial statements. To ensure data quality, we require non-missing values for both ACOX and ICAPT, and we exclude observations where invested capital equals zero to avoid undefined ratios. The resulting signal measures the intensity of miscellaneous current asset deployment relative to the firm’s total invested capital base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the MCI signal. Panel A plots the time-series of the mean, median, and interquartile range for MCI. On average, the cross-sectional mean (median) MCI is 0.03 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input MCI data. The signal’s interquartile range spans -0.00 to 0.07. Panel B of Figure 1 plots the time-series of

the coverage of the MCI signal for the CRSP universe. On average, the MCI signal is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

4 Does MCI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on MCI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high MCI portfolio and sells the low MCI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short MCI strategy earns an average return of 0.20% per month with a t-statistic of 2.37. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.26 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.48, with a t-statistic of -14.58 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 634 stocks and an average market capitalization of at least \$1,489 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.80. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-19bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the MCI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in

nineteen cases.

Table 3 provides direct tests for the role size plays in the MCI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and MCI, as well as average returns and alphas for long/short trading MCI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the MCI strategy achieves an average return of 21 bps/month with a t-statistic of 2.26. Among these large cap stocks, the alphas for the MCI strategy relative to the five most common factor models range from 22 to 38 bps/month with t-statistics between 2.26 and 4.75.

5 How does MCI perform relative to the zoo?

Figure 2 puts the performance of MCI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the MCI strategy falls in the distribution. The MCI strategy’s gross (net) Sharpe ratio of 0.30 (0.27) is greater than 65% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the MCI strategy (red line).² Ignoring trading costs, a \$1 invested in the MCI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.93 which ranks the MCI strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the MCI strategy would have yielded \$2.31 which ranks the MCI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the MCI relative to those. Panel A shows that the MCI strategy gross alphas fall between the 41 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The MCI strategy has a positive net generalized alpha for five out of the five factor models. In these cases MCI ranks between the 65 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does MCI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of MCI with 210 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price MCI or at least to weaken the power MCI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI}MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on MCI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the MCI signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.90, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on MCI is 2.91.

Similarly, Table 5 reports results from spanning tests that regress returns to the MCI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the MCI strategy earns alphas that range from 26-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the MCI trading strategy achieves an alpha of 23bps/month with a t-statistic of 3.60.

7 Does MCI add relative to the whole zoo?

Finally, we can ask how much adding MCI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the MCI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes MCI grows to \$3431.66.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which MCI is available.

8 Conclusion

This study provides robust evidence that Miscellaneous Capital Intensity (MCI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that MCI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.30 gross (0.27 net), indicating strong performance that remains economically viable after accounting for transaction costs. Most notably, the strategy delivers statistically significant alphas of 29 basis points per month (t-statistic = 4.02) relative to the Fama-French five-factor model plus momentum, and maintains a significant alpha of 23 basis points per month (t-statistic = 3.60) even when controlling for six closely related strategies from the factor zoo. These findings suggest that MCI captures unique information about firm fundamentals not fully reflected in existing pricing models or related value and leverage metrics. The economic magnitude and statistical robustness of the MCI signal have important practical implications for portfolio construction and risk management. Asset managers can potentially enhance portfolio performance by incorporating MCI into their investment processes, while the signal’s persistence after transaction costs suggests implementability in real-world trading environments. Furthermore, the independence of MCI from related factors indicates opportunities for diversification benefits when combined with existing factor strategies. Several limitations of this study merit consideration and point toward productive avenues for future research. First, while our analysis demonstrates the predictive power of MCI in U.S. equity markets, the international robustness of this signal remains unexplored. Future research should examine whether MCI’s predictive ability extends to developed and emerging markets globally. Second, the economic mechanisms underlying the MCI anomaly warrant

deeper investigation. Understanding whether the returns to MCI reflect compensation for systematic risk, market frictions, or behavioral biases would enhance our theoretical understanding of this phenomenon. Third, the time-varying nature of the MCI premium deserves attention, particularly its behavior across different market regimes and business cycles. Finally, future work could explore potential enhancements to the MCI signal through machine learning techniques or by combining it with other firm characteristics in more sophisticated ways. Despite these limitations, our findings establish MCI as a robust and economically significant predictor of stock returns that adds value beyond existing factors and warrants inclusion in the canon of documented return anomalies.

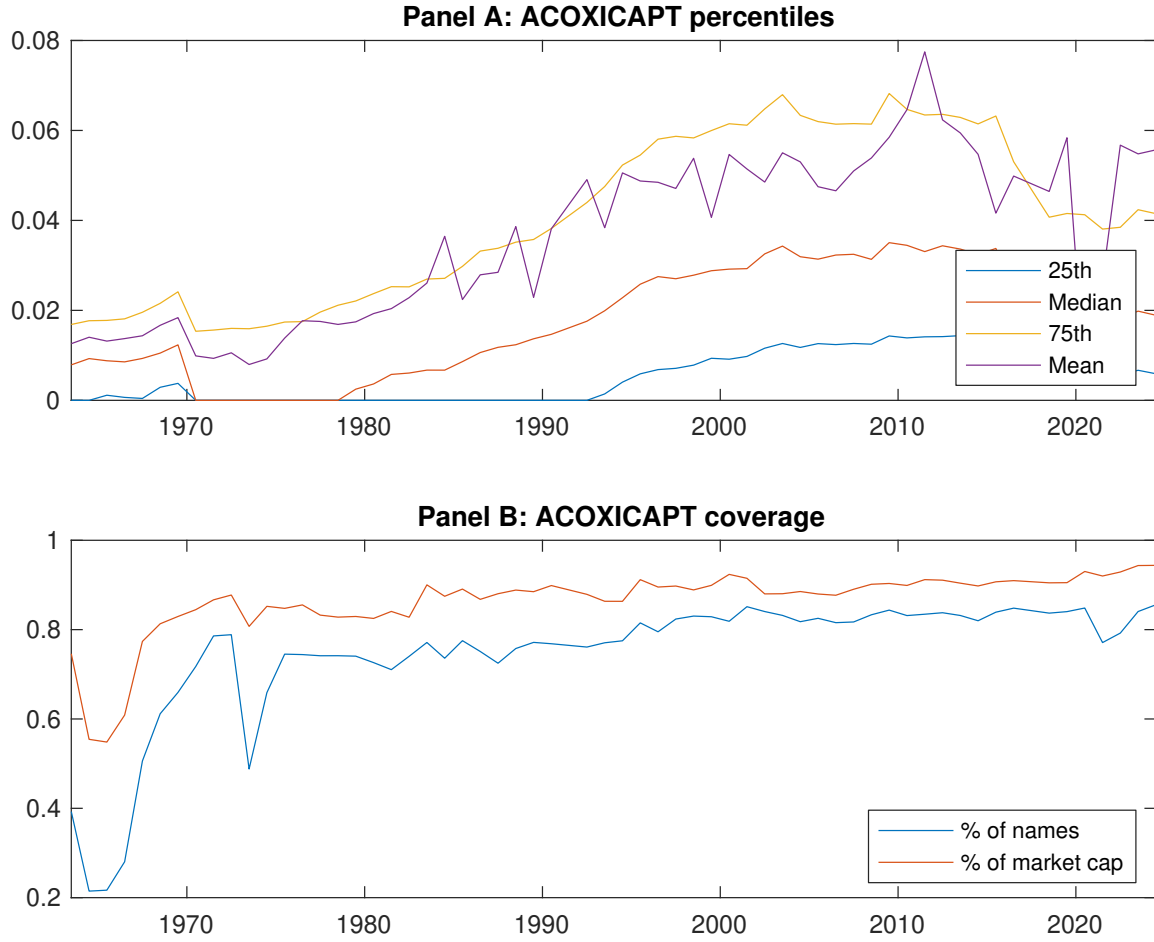


Figure 1: Times series of MCI percentiles and coverage. This figure plots descriptive statistics for MCI. Panel A shows cross-sectional percentiles of MCI over the sample. Panel B plots the monthly coverage of MCI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on MCI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on MCI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [3.08]	0.55 [3.11]	0.62 [3.57]	0.60 [3.50]	0.74 [4.24]	0.20 [2.37]
α_{CAPM}	-0.05 [-0.93]	-0.06 [-1.27]	0.02 [0.53]	0.01 [0.26]	0.14 [3.06]	0.19 [2.26]
α_{FF3}	-0.14 [-3.08]	-0.07 [-1.45]	0.05 [1.08]	0.07 [1.57]	0.21 [4.92]	0.35 [5.07]
α_{FF4}	-0.13 [-2.75]	-0.05 [-1.08]	0.03 [0.74]	0.06 [1.42]	0.19 [4.50]	0.32 [4.60]
α_{FF5}	-0.16 [-3.40]	-0.06 [-1.21]	0.03 [0.59]	0.08 [1.79]	0.14 [3.46]	0.31 [4.34]
α_{FF6}	-0.15 [-3.11]	-0.04 [-0.91]	0.02 [0.35]	0.07 [1.63]	0.14 [3.24]	0.29 [4.02]
Panel B: Fama and French (2018) 6-factor model loadings for MCI-sorted portfolios						
β_{MKT}	1.06 [90.80]	1.03 [89.44]	1.00 [87.20]	0.97 [91.70]	1.01 [100.94]	-0.05 [-3.09]
β_{SMB}	-0.07 [-4.35]	-0.00 [-0.15]	0.03 [1.91]	0.02 [1.55]	0.05 [3.30]	0.12 [4.90]
β_{HML}	0.27 [12.04]	0.01 [0.43]	-0.08 [-3.72]	-0.14 [-6.69]	-0.21 [-10.90]	-0.48 [-14.58]
β_{RMW}	0.06 [2.49]	-0.03 [-1.47]	0.05 [2.16]	-0.01 [-0.38]	0.16 [8.00]	0.10 [2.97]
β_{CMA}	0.00 [0.04]	0.02 [0.62]	0.02 [0.67]	-0.05 [-1.51]	0.05 [1.77]	0.05 [1.01]
β_{UMD}	-0.02 [-1.46]	-0.02 [-1.70]	0.02 [1.40]	0.01 [0.81]	0.01 [1.00]	0.03 [1.58]
Panel C: Average number of firms (n) and market capitalization (me)						
n	688	728	690	639	634	
me (\$10 ⁶)	1489	1519	1872	2157	2444	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the MCI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [2.37]	0.19 [2.26]	0.35 [5.07]	0.32 [4.60]	0.31 [4.34]	0.29 [4.02]
Quintile	NYSE	EW	0.22 [3.92]	0.18 [3.22]	0.20 [3.84]	0.20 [3.70]	0.19 [3.45]	0.19 [3.38]
Quintile	Name	VW	0.21 [2.59]	0.21 [2.48]	0.36 [5.33]	0.34 [4.89]	0.32 [4.56]	0.30 [4.26]
Quintile	Cap	VW	0.20 [2.44]	0.20 [2.34]	0.34 [4.89]	0.34 [4.69]	0.31 [4.36]	0.31 [4.24]
Decile	NYSE	VW	0.18 [1.80]	0.18 [1.76]	0.33 [3.73]	0.33 [3.72]	0.28 [3.10]	0.29 [3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [2.10]	0.18 [2.12]	0.32 [4.61]	0.30 [4.38]	0.27 [3.90]	0.26 [3.74]
Quintile	NYSE	EW	0.05 [0.81]	0.01 [0.10]	0.02 [0.39]	0.02 [0.41]		
Quintile	Name	VW	0.19 [2.30]	0.19 [2.26]	0.33 [4.78]	0.31 [4.56]	0.28 [4.03]	0.27 [3.89]
Quintile	Cap	VW	0.18 [2.13]	0.17 [1.96]	0.30 [4.19]	0.29 [4.10]	0.25 [3.58]	0.25 [3.54]
Decile	NYSE	VW	0.15 [1.48]	0.15 [1.48]	0.28 [3.21]	0.28 [3.24]	0.23 [2.59]	0.23 [2.64]

Table 3: Conditional sort on size and MCI

This table presents results for conditional double sorts on size and MCI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on MCI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high MCI and short stocks with low MCI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	MCI Quintiles					MCI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.54 [2.24]	0.66 [2.56]	0.79 [3.10]	0.80 [3.14]	0.88 [3.21]	0.34 [2.97]	0.29 [2.52]	0.29 [2.54]	0.27 [2.34]	0.28 [2.41]	0.27 [2.28]
	(2)	0.70 [3.12]	0.72 [3.03]	0.80 [3.30]	0.80 [3.33]	0.90 [3.76]	0.19 [2.28]	0.16 [1.88]	0.19 [2.32]	0.20 [2.37]	0.19 [2.20]	0.20 [2.28]
	(3)	0.71 [3.41]	0.69 [3.29]	0.77 [3.49]	0.81 [3.62]	0.91 [4.07]	0.20 [2.41]	0.15 [1.85]	0.19 [2.43]	0.21 [2.63]	0.22 [2.75]	0.24 [2.90]
	(4)	0.66 [3.34]	0.57 [2.84]	0.69 [3.50]	0.85 [4.14]	0.86 [4.11]	0.19 [2.12]	0.16 [1.78]	0.26 [3.07]	0.25 [2.91]	0.25 [2.92]	0.24 [2.81]
	(5)	0.50 [2.86]	0.52 [3.10]	0.60 [3.49]	0.58 [3.48]	0.71 [4.17]	0.21 [2.26]	0.22 [2.26]	0.38 [4.75]	0.36 [4.43]	0.34 [4.15]	0.33 [3.94]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	MCI Quintiles					MCI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	382	386	388	387	381	29	32	33	32	28	
	(2)	104	104	104	104	104	53	54	54	54	54	
	(3)	73	72	73	72	73	92	92	92	92	93	
	(4)	60	60	60	60	60	197	193	196	200	202	
(5)	55	55	55	55	55	1349	1283	1555	1633	1789		

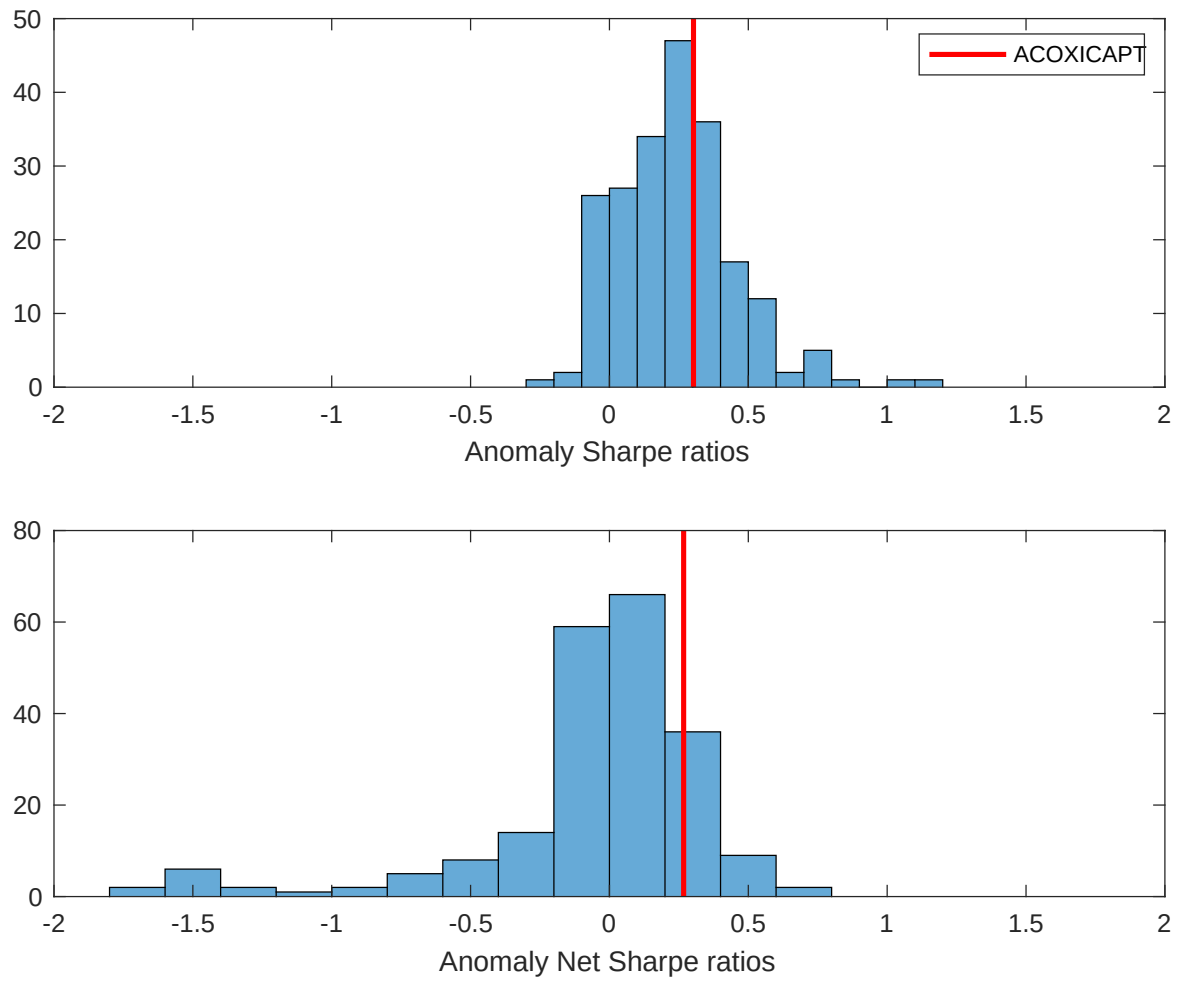


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the MCI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

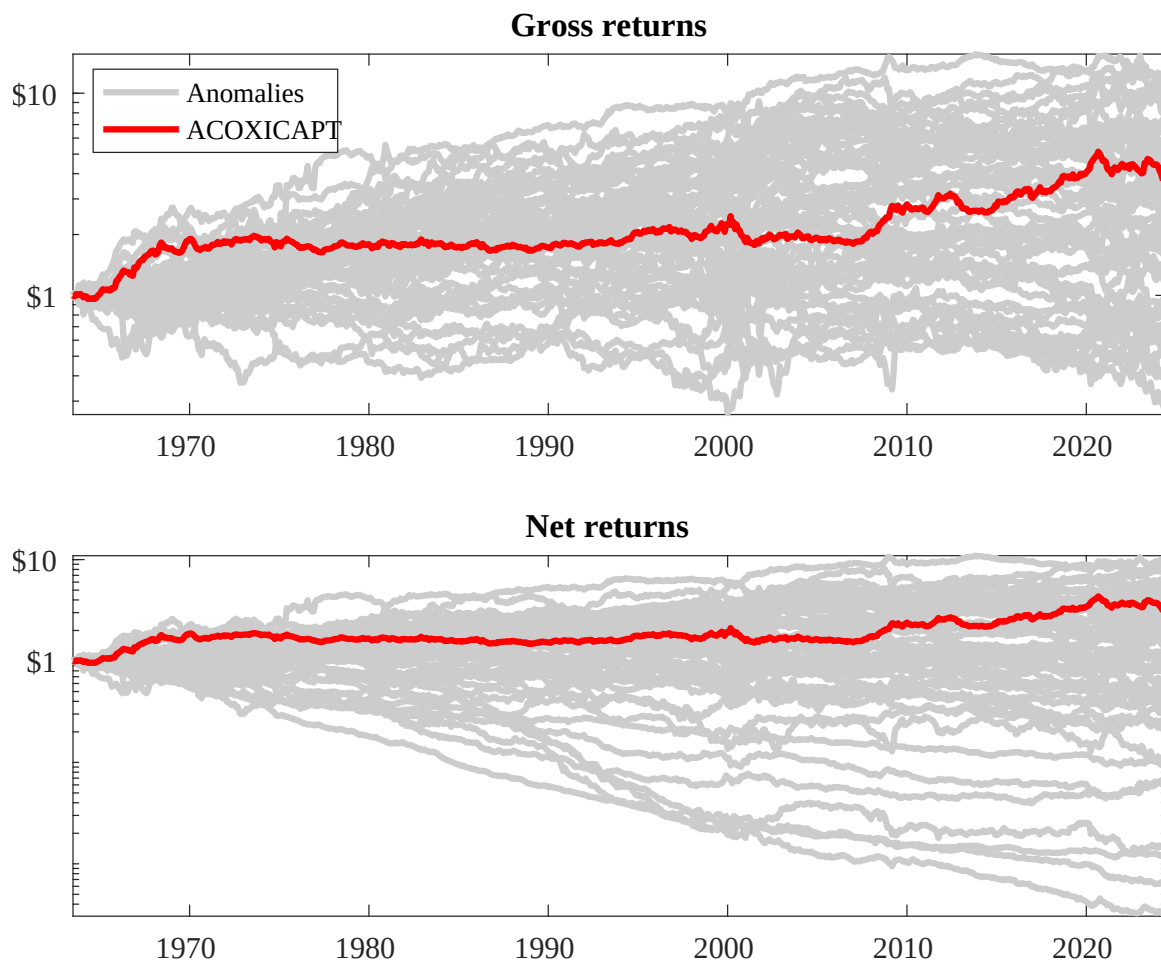
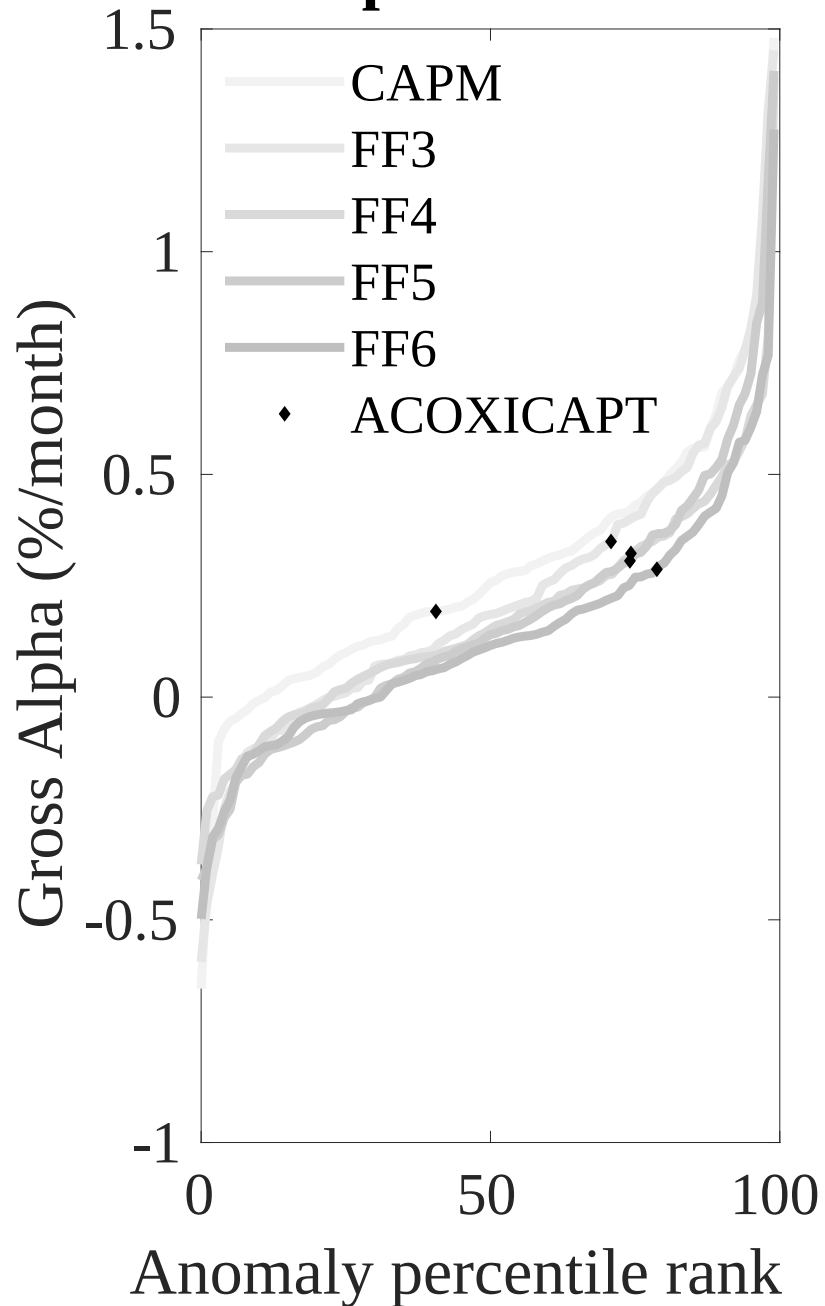


Figure 3: Dollar invested.

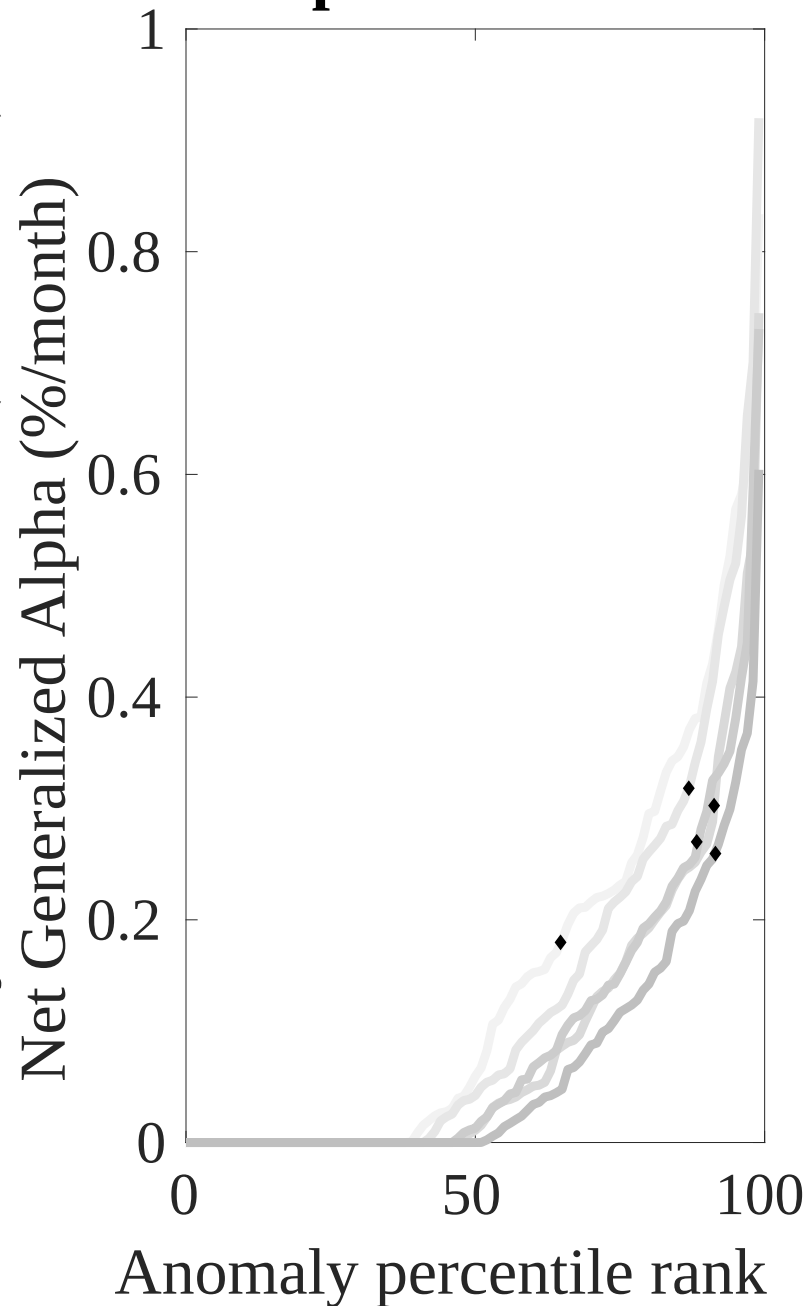
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the MCI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



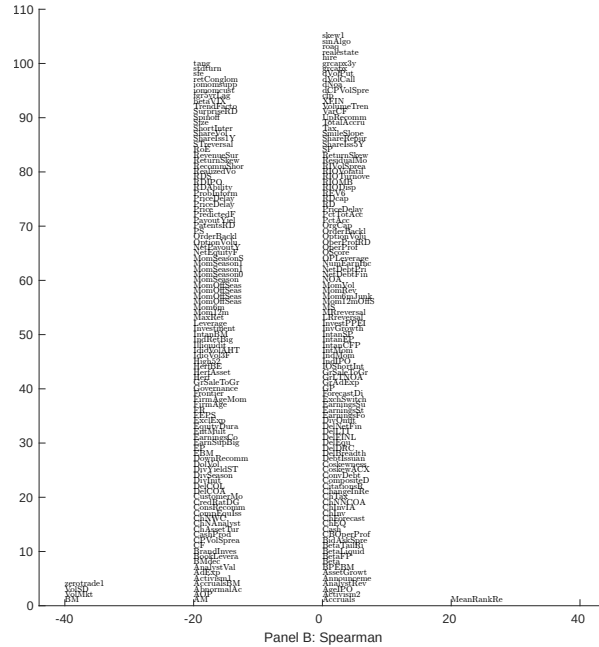
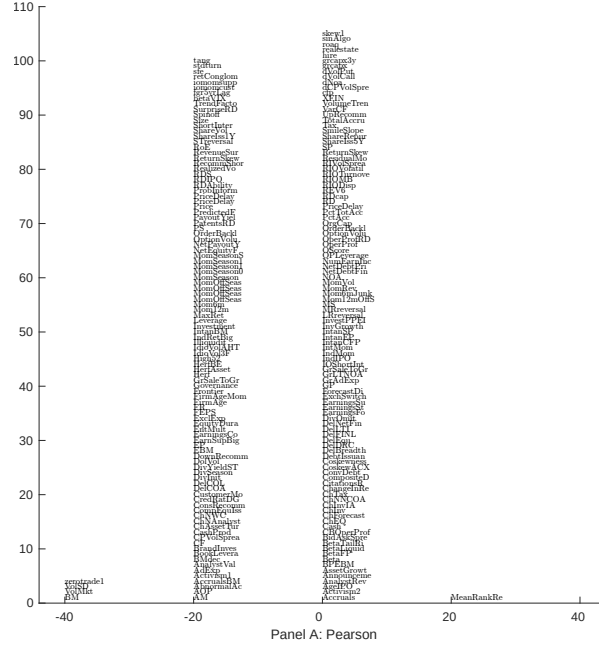


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with MCI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

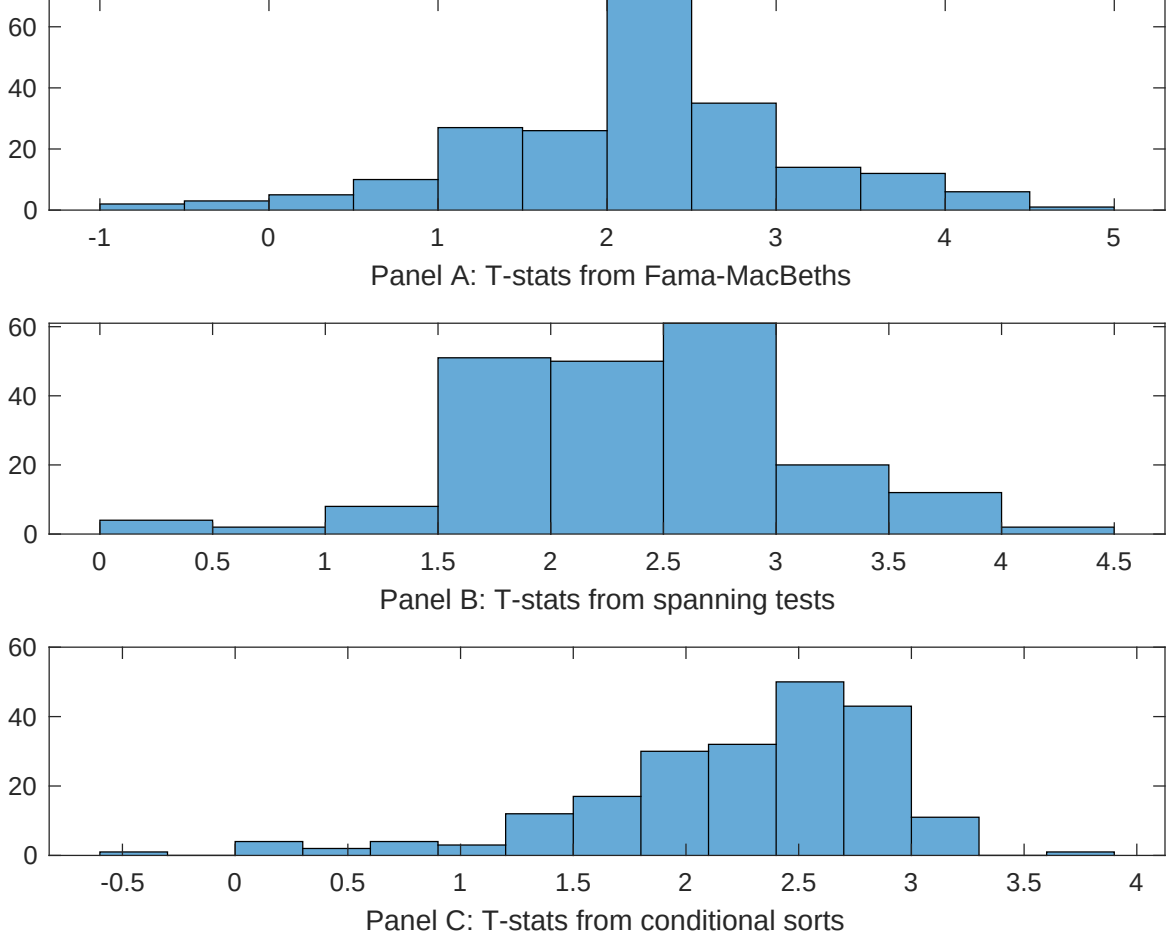


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI}MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on MCI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{MCI}MCI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.24]	0.87 [3.71]	0.20 [9.05]	0.10 [4.48]	0.10 [4.85]	0.99 [4.55]	0.22 [7.87]
MCI	0.21 [3.52]	0.19 [3.09]	0.16 [2.93]	0.10 [1.90]	0.16 [3.03]	0.13 [2.30]	0.14 [2.91]
Anomaly 1	0.38 [7.83]						0.94 [2.17]
Anomaly 2		0.35 [5.40]					-0.10 [-1.34]
Anomaly 3			0.53 [5.42]				0.67 [5.97]
Anomaly 4				0.11 [3.05]			0.10 [3.09]
Anomaly 5					0.17 [2.64]		-0.20 [-0.31]
Anomaly 6						0.58 [3.38]	0.12 [0.65]
# months	727	727	720	732	727	727	715
$\bar{R}^2(\%)$	1	1	1	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the MCI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{MCI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Book to market, original (Statman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27 [3.90]	0.27 [3.89]	0.29 [4.06]	0.26 [3.85]	0.30 [4.27]	0.28 [4.18]	0.23 [3.60]
Anomaly 1	-25.12 [-6.24]						-16.63 [-3.39]
Anomaly 2		-21.76 [-4.87]					-11.54 [-2.18]
Anomaly 3			-11.68 [-3.21]				7.55 [1.72]
Anomaly 4				23.46 [8.61]			14.30 [5.04]
Anomaly 5					-9.48 [-3.04]		2.38 [0.69]
Anomaly 6						-29.77 [-10.53]	-22.83 [-7.27]
mkt	-4.26 [-2.56]	-4.79 [-2.86]	-5.00 [-2.96]	-4.91 [-3.03]	-5.46 [-3.22]	-1.01 [-0.62]	-1.10 [-0.69]
smb	22.18 [7.86]	20.20 [7.09]	16.21 [6.09]	2.31 [0.87]	14.82 [5.87]	16.89 [7.27]	16.92 [5.55]
hml	-21.06 [-4.05]	-24.22 [-4.31]	-33.95 [-6.50]	-34.65 [-10.05]	-37.57 [-8.52]	-12.23 [-2.73]	5.32 [0.88]
rmw	5.89 [1.79]	4.73 [1.38]	9.32 [2.83]	-3.30 [-0.94]	11.19 [3.36]	6.27 [2.02]	-6.22 [-1.74]
cma	5.88 [1.25]	5.06 [1.06]	1.68 [0.35]	-0.54 [-0.12]	2.95 [0.61]	-1.96 [-0.43]	0.49 [0.11]
umd	4.09 [2.48]	3.61 [2.17]	2.79 [1.67]	2.46 [1.53]	1.30 [0.74]	-5.45 [-3.09]	-2.16 [-1.18]
# months	728	728	732	732	728	728	728
$\bar{R}^2(\%)$	39	38	37	42	36	44	48

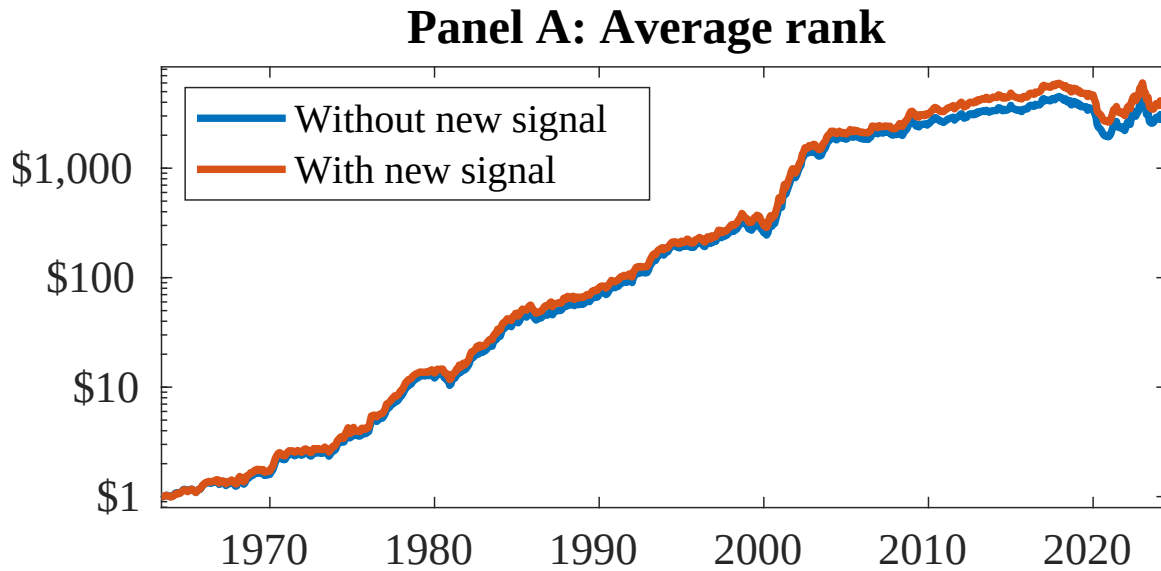


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as MCI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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